GRADE 11 TECHPRO ACTIVITY SHEET
CSS NAVIGATION BARS

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SECTION: 11 - Bell

Learning Objectives

- Explain the purpose of a website navigation bar.
- Build a navigation bar using HTML <nav>, <ul>, <li>, and <a> elements.
- Style a vertical navigation bar using display: block, padding, and hover effects.
- Style a horizontal navigation bar using float or display: flex.
- Highlight the active link and improve navigation readability.
- Build and test a simple TechPro webpage in Visual Studio Code.

Instructions

- Create a folder named CSS_Navbar_Activity and open it in Visual Studio Code.
- Create a file named index.html.
- Enter the starter code, save it, and open it in a browser. Live Server is optional.
- Complete each part in order. Change values and refresh the browser to observe the results.
- Keep a backup copy before trying the challenge task.

PART A — Create the Starter Page

Enter this complete starter code in index.html. You will add styles and practice examples in the following parts.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Navigation Bar Activity</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background: #f7f3f4;
      color: #222;
    }
    header {
      padding: 24px;
      background: maroon;
      color: white;
    }
    main { padding: 24px; }
  </style>
</head>
<body>
  <header>
    <h1>My TechPro Website</h1>
    <p>Practicing CSS Navigation Bars</p>
  </header>
  <main>
    <h2>Welcome!</h2>
    <p>This page will contain a vertical and a horizontal navigation bar.</p>
```

```
  </main>
</body>
</html>
```

PART B — Navigation Bar Basics

A navigation bar is a list of links that helps visitors move between pages or sections. The HTML structure commonly uses <nav>, <ul>, <li>, and <a>. Add this inside <body>, before <main>:

```
<nav aria-label="Main navigation">
  <ul>
    <li><a href="#home">Home</a></li>
    <li><a href="#lessons">Lessons</a></li>
    <li><a href="#projects">Projects</a></li>
    <li><a href="#about">About</a></li>
  </ul>
</nav>
```

Add this base CSS inside <style>. It removes the default bullets, margin, and padding from the list.

```
nav ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
}
```

PART C — Create a Vertical Navigation Bar

A vertical navigation bar stacks links one below another, often along the side of a page. Add a class to the nav element: <nav class="vertical-nav">. Then use this CSS:

```
 .vertical-nav ul {
  width: 200px;
  background-color: #eeeeee;
}
.vertical-nav a {
  display: block;
  color: #222222;
  padding: 10px 16px;
  text-decoration: none;
}
.vertical-nav a:hover {
  background-color: #555555;
  color: white;
}
.vertical-nav a.active {
  background-color: maroon;
  color: white;
}
```

Activity:

- Set the navigation link Home to class="active".
- Change the width from 200px to 240px and compare.
- Change the hover background to dark red or navy.
- Add a border around the navigation list.

PART D — Create a Horizontal Navigation Bar

A horizontal navigation bar places links next to each other, commonly near the top of a webpage. For this part, use a separate nav element with class="horizontal-nav" and the same link labels.

```
<nav class="horizontal-nav" aria-label="Horizontal navigation">
  <ul>
```

```
    <li><a class="active" href="#home">Home</a></li>
    <li><a href="#lessons">Lessons</a></li>
    <li><a href="#projects">Projects</a></li>
    <li><a href="#about">About</a></li>
  </ul>
</nav>
```

Add this CSS. It uses Flexbox to place the links in a row.

```
 .horizontal-nav ul {
  display: flex;
  background-color: #333333;
}
.horizontal-nav a {
  display: block;
  color: white;
  padding: 14px 18px;
  text-decoration: none;
  text-align: center;
}
.horizontal-nav a:hover {
  background-color: #111111;
}
.horizontal-nav a.active {
  background-color: maroon;
}
```

Activity:

- Add justify-content: center; to the horizontal list.
- Change the bar background from #333333 to maroon.
- Change the padding and observe how the clickable area changes.
- Try display: inline; on list items in a separate experiment, then compare it with Flexbox.

PART E — Link States and Borders

The :hover pseudo-class styles a link while the pointer is over it. An active class can mark the current page or section. Add a divider between horizontal links if you wish.

```
 .horizontal-nav li {
  border-right: 1px solid #777777;
}
.horizontal-nav li:last-child {
  border-right: none;
}
```

Task: Make sure the active link is easy to identify. Test every link and confirm that it jumps to the matching section.

PART F — Connect Links to Page Sections

Create matching section IDs so the links work within the same page. Add these sections inside <main>:

```
<section id="home">
  <h2>Home</h2>
  <p>Welcome to our TechPro learning page.</p>
</section>
<section id="lessons">
  <h2>Lessons</h2>
  <p>We practice HTML and CSS in Visual Studio Code.</p>
</section>
<section id="projects">
  <h2>Projects</h2>
  <p>We create simple webpages and design activities.</p>
</section>
<section id="about">
  <h2>About</h2>
```

```
  <p>This page is a CSS navigation practice activity.</p>
</section>
```

Note: Use each section ID only once on the page. Keep the IDs in href values exactly the same as the section IDs.

PART G — Mini Project: TechPro Learning Portal

Build a one-page "TechPro Learning Portal" that includes all requirements below:

- A maroon header with the website title.
- A horizontal navigation bar with at least four links.
- A vertical navigation bar or sidebar with at least three links.
- Hover effects that change link appearance.
- An active link style.
- Links that jump to matching page sections.
- Readable text, clear spacing, and strong color contrast.

Suggested styling ideas: use maroon, white, light gray, and dark gray. You may customize the colors and layout as long as the navigation remains clear.

PART H — Challenge Task

Create a second navigation design using your own theme, such as a school library, ICT club, student portfolio, or online class page.

- Use a horizontal menu at the top and a vertical menu beside the content.
- Add a visible hover effect and active link.
- Add borders or spacing to separate menu items.
- Explain in two or three sentences when a vertical menu may be more useful than a horizontal menu.

Check Your Work

☐ Created CSS_Navbar_Activity and index.html.

☐ Opened the webpage in a browser.

☐ Used nav, ul, li, and a elements.

☐ Removed default list markers, margins, and padding.

☐ Created a vertical navigation bar.

☐ Created a horizontal navigation bar using Flexbox.

☐ Added hover and active styles.

☐ Connected links to matching section IDs.

☐ Completed the TechPro Learning Portal mini-project.

Review Questions

1. What is the purpose of a navigation bar?

Answer: to have a list of links used to move around a website or page

2. Which HTML elements are commonly used to build a navigation bar?

Answer: <nav>, <ul>, <li>, and <a> elements

3. What does list-style-type: none do in a navigation list?

Answer: removes lists markers

4. Why is display: block useful for links in a vertical navigation bar?

Answer: makes the link area fill its available row and easier to click

5. What is the difference between a vertical and horizontal navigation bar?

Answer: a vertical navbar has links stacked from top to bottom, while a horizontal navbar arranges links in a row

6. What does the :hover pseudo-class do?

Answer: applies a style when the pointer is over a link

7. How does the active class help users?

Answer: It can mark the current page or section the user is in

8. What does display: flex do in the horizontal navbar example?

Answer: arranges items in a row by default

9. Why should an href such as #lessons match a section id="lessons"?

Answer: because the IDs in href values must be kept exactly the same as the section IDs

10. Name one design choice that improves navigation readability.

Answer: readable text, clear spacing, and good color contrast

Quick Review

Navigation bar = a list of links used to move around a website or page.

Vertical navbar = links stacked from top to bottom.

Horizontal navbar = links arranged in a row.

list-style-type: none; = removes list markers.

display: block; = makes the link area fill its available row and easier to click.

:hover = applies a style when the pointer is over a link.

display: flex; = arranges items in a row by default.

References

W3Schools — CSS Navigation Bar: https://www.w3schools.com/css/css_navbar.asp

W3Schools — CSS Vertical Navigation Bar: https://www.w3schools.com/css/css_navbar_vertical.asp

W3Schools — CSS Horizontal Navigation Bar: https://www.w3schools.com/css/css_navbar_horizontal.asp